OCR GCSE Geography B (J384) — Paper 1: Our Natural World · Topic 4: Sustaining Ecosystems

1.6 Sustaining Ecosystems: Tropical Rainforests

1. Biomes and Global Climate Zones

A **biome** is a very large scale ecosystem, defined by its climate and the plant and animal communities that climate supports. Biomes are not scattered at random: their distribution is controlled almost entirely by the **global circulation of the atmosphere**, which fixes where air rises (low pressure, wet) and where it sinks (high pressure, dry). Because that circulation is arranged in belts either side of the equator, the world's biomes also form broad latitudinal belts.

> **Biome:** A global scale ecosystem in which the climate, soils, vegetation and animal life are closely interdependent — for example tropical rainforest, hot semi-arid grassland (savanna), tundra or temperate deciduous forest.

How Global Circulation Fixes Biome Distribution

Latitude	Air movement	Pressure and rainfall	Resulting biome
0° (equator)	Rising air at the ITCZ	Low pressure; rain all year, over 2,000 mm	Tropical rainforest
5–20° N and S	ITCZ migrates north and south with the overhead sun	Alternating low and high pressure; a wet season and a long dry season	Hot semi-arid grassland (savanna)
20–30° N and S	Descending limb of the Hadley cell	Permanent high pressure; under 250 mm	Hot desert
40–60° N	Rising air where polar and tropical air meet	Low pressure; rain spread through the year	Temperate deciduous forest

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60–70° N	Cold, dry, sinking air	High pressure; low precipitation, mostly snow	Tundra

OCR B Topic 4 asks you to understand what ecosystems are and where the world's major biomes are found, and to study two very different environments in depth: **tropical rainforests** (this topic) and **polar environments** (topic 1.7), including how each can be managed sustainably.

Exam Tip — Describing Distribution: When asked to describe where a biome is found, always name the latitude band, name two or three specific regions or countries, and link the pattern to pressure. "Savanna is found roughly between 5° and 20° north and south of the equator — across the Sahel, East Africa, northern Australia and the Brazilian cerrado — where the ITCZ brings rain for part of the year and high pressure dominates for the rest." That single sentence scores far more than "it is found in hot places".

2. Energy Flows and Food Webs

Nutrient cycling moves matter round and round an ecosystem. **Energy does not cycle** — it enters as sunlight, passes along a one-way chain, and is lost as heat. Understanding that difference is the point of this part of the specification.

Producer: A green plant that converts sunlight into chemical energy by photosynthesis. Producers form the base of every food web.
Consumer: An animal that obtains energy by eating other organisms — primary consumers eat plants, secondary consumers eat primary consumers, tertiary consumers eat secondary consumers.
Decomposer: A bacterium, fungus or detritivore that breaks down dead material and returns nutrients to the soil.

Trophic Levels

Each step in a food chain is a **trophic level**. Only about **10% of the energy** at one level is passed on to the next — the rest is used in respiration, movement and body heat, or is lost in waste. This is why food chains rarely have more than four or five links, and why there are always far fewer top predators than producers.

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Trophic level	Tropical rainforest	Hot semi-arid grassland
Producers	Canopy trees, lianas, epiphytes, forest-floor ferns	Red oat grass, elephant grass, acacia, baobab
Primary consumers	Leaf-cutter ants, sloths, howler monkeys, tapir	Wildebeest, zebra, gazelle, elephant, termites
Secondary consumers	Poison dart frogs, tree snakes, army ants	Jackal, serval, secretary bird
Tertiary consumers	Jaguar, harpy eagle, anaconda	Lion, cheetah, spotted hyena
Decomposers	Fungi, bacteria, termites — working within days in the constant warmth and damp	Termites and dung beetles in the wet season; decomposition almost stops in the dry season

Why the Two Biomes Differ

- **Rainforest — energy concentrated at the base.** Constant warmth, light and water mean net primary productivity is the highest of any land biome. Most of the biomass sits in the standing trees, and most consumers are small-bodied specialists living in the canopy.
- **Savanna — energy concentrated in large herbivores.** Productivity is lower and strongly seasonal, but grass is far easier to eat than wood, so an unusually large share of the plant growth is converted into herbivore biomass. This is why the savanna, not the rainforest, carries the world's great herds.
- **Decomposition speed.** In the rainforest, leaf litter is broken down in weeks and nutrients are taken straight back up by shallow roots. In the savanna, dead grass lies dry and unrotted for months, which is why fire, rather than decomposition, is often what recycles it.

Biodiversity and Food Web Stability

A food web with many species at each level is more **resilient**: if one prey species declines, predators can switch to another, and the web holds. A simplified web — for example one created when farming replaces natural vegetation with a single crop — has no such alternatives, so the loss of one species can collapse several others. This is the mechanism behind the damage described in 3.2.

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Exam Tip — Drawing and Reading Food Webs: Arrows in a food web always point in the direction that *energy flows*, i.e. from the organism being eaten to the organism eating it. A common and costly error is to draw them the other way. If asked to predict the effect of removing a species, trace every arrow that touches it and describe both the level above (loses a food source) and the level below (population rises).

3. Location and Climate

Tropical rainforests are found between the **Tropic of Cancer (23.5°N) and Tropic of Capricorn (23.5°S)**, with the densest concentration within 5° either side of the equator. Their location is driven by the **Inter-Tropical Convergence Zone (ITCZ)** — a belt of low pressure where trade winds meet, causing warm, moist air to rise, cool, and produce intense rainfall.

Climate Characteristics

Variable	Value	Detail
Mean annual temperature	~27°C	Very low seasonal variation (1–2°C). Hot all year due to direct overhead sun.
Annual rainfall	>2,000 mm/year	Rainfall distributed throughout the year; convectional rainfall common in afternoons.
Humidity	Very high (80–90%)	Constant evapotranspiration from dense vegetation keeps air saturated.
Seasonality	None / very slight	No true dry season at the equator; ITCZ movement creates slight wetter/drier periods.

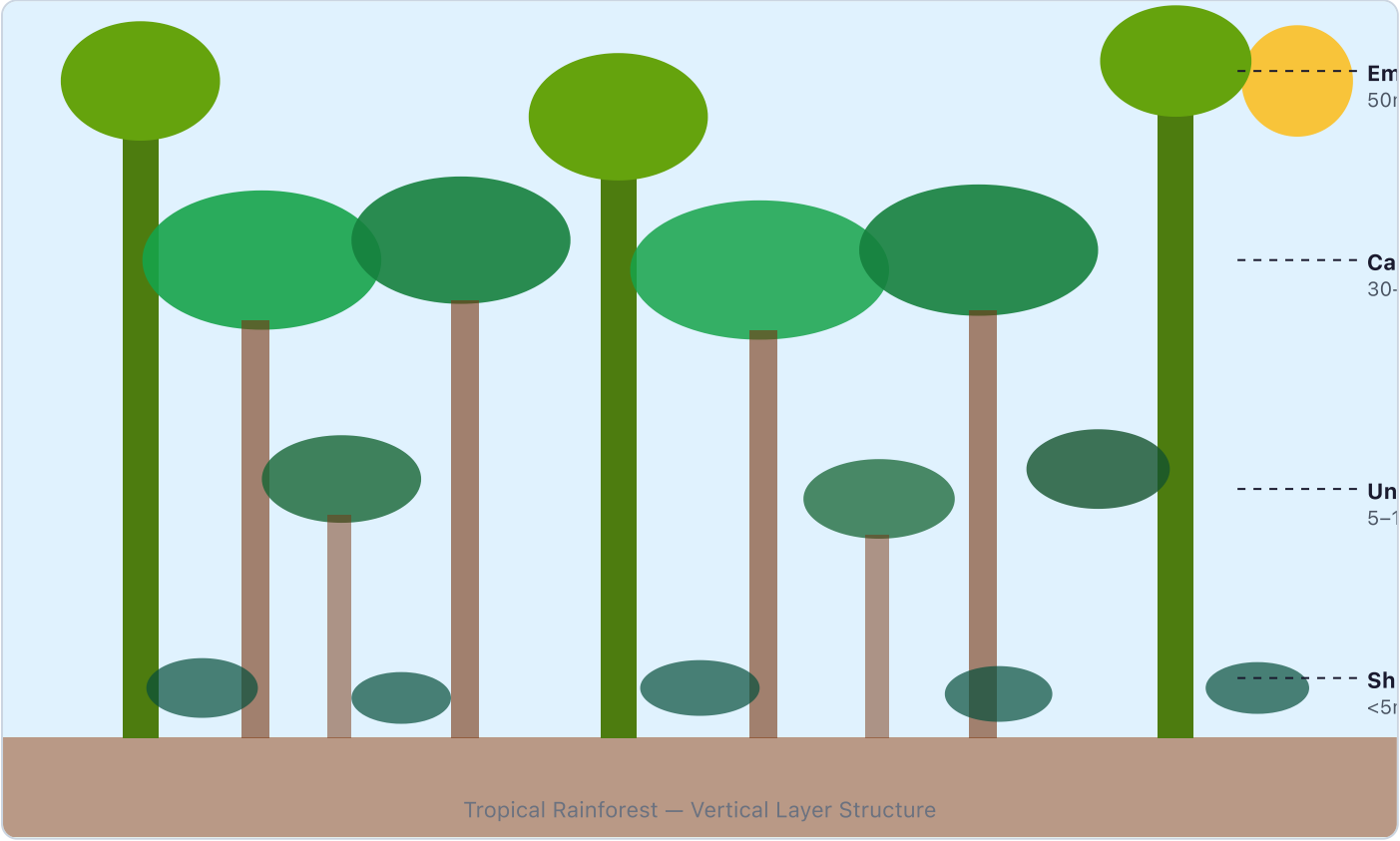
Major Tropical Rainforest Areas

Region	Location	Extent
Amazon Basin	South America (mainly Brazil)	~5.5 million km² — largest in the world

Congo Basin	Central Africa (DRC, Congo, Cameroon)	~1.7 million km² — second largest
Borneo / SE Asian forests	Indonesia, Malaysia, Papua New Guinea	Highly threatened; orangutan habitat

4. Structure of the Rainforest

The rainforest has a **stratified structure** — four distinct vertical layers, each with different light levels, temperature, and wildlife. This layering creates dozens of different microhabitats supporting extraordinary biodiversity.



Layer	Height	Light Level	Plants	Animals
Emergent	50m+	Full sunlight	Kapok, Brazil nut tree — isolated giant trees with umbrella-shaped	Harpy eagles, macaws, butterflies

			crowns	
Canopy	30–40m	~80% of light	Dense, interlocking branches forming a continuous roof; where 80% of photosynthesis occurs	Monkeys, toucans, tree frogs, sloths
Understorey	5–15m	Very low (~2–5%)	Adapted to shade; large leaves to maximise light absorption; young canopy trees	Jaguars, snakes, insects
Shrub/Ground	<5m	Less than 1%	Sparse; mosses, ferns, fungi, seedlings that wait for a gap in canopy	Tapirs, ants, decomposers, gorillas (Congo)

Exam Tip: When describing rainforest structure, always link the layer to its light level and explain how plants in that layer are adapted to survive with that amount of light. This shows geographical understanding rather than just recall.

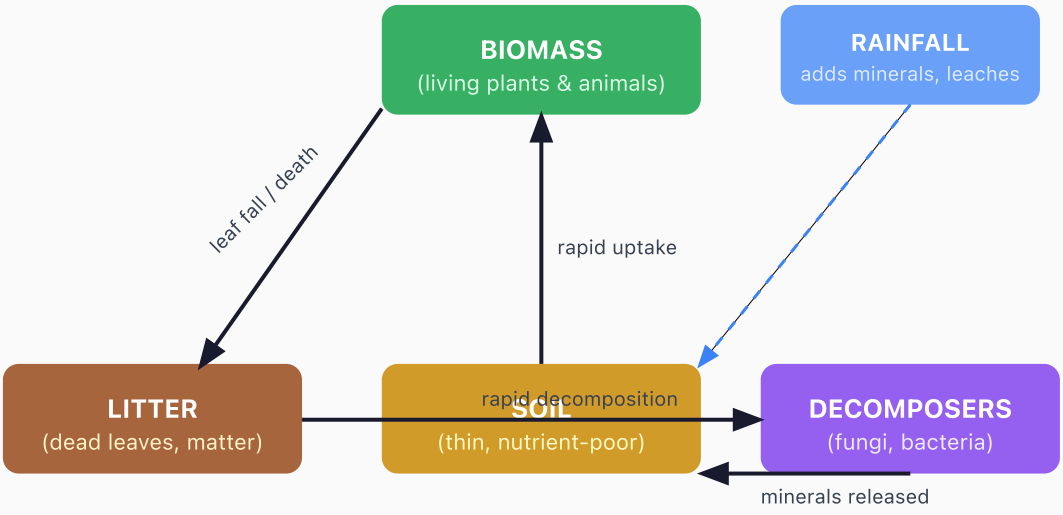
5. Nutrient Cycling

The tropical rainforest has a **rapid nutrient cycle**, driven by the hot, wet climate. Despite the lush vegetation, the soils are actually **thin and nutrient-poor (latosols/oxisols)** — almost all nutrients are locked up in living biomass (plants and animals).

How the Nutrient Cycle Works

Dead leaves and plant matter (litter) fall to the forest floor. The warm, moist conditions mean **decomposers (fungi and bacteria) break down this litter extremely quickly** — sometimes within weeks. Nutrients are released into the thin topsoil and immediately taken up by the shallow root systems of trees. The cycle is therefore almost closed — nutrients flow rapidly from litter to decomposers to soil to plants and back again.

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Additional Nuance: The Role of Mycorrhizal Networks

A critical but often overlooked component of the nutrient cycle is the **mycorrhizal fungal network** — a web of fungal threads (hyphae) that connects the roots of different trees underground. These fungi form mutually beneficial (mutualistic) relationships with tree roots: the tree provides the fungus with sugars produced by photosynthesis, and in return the fungus dramatically extends the tree's ability to absorb phosphorus, nitrogen, and water from the soil. This means nutrients can be transferred between trees through the fungal network, making the cycle even more efficient and resilient. In old-growth rainforest, up to 90% of nutrient uptake may be mediated by mycorrhizal fungi.

Why Leaching is Such a Threat

The intense rainfall (2,000+ mm/year) constantly percolates through the soil, dissolving and washing nutrients downward in a process called **leaching**. This is why the nutrient cycle must be so fast — any nutrients that are not taken up almost immediately by roots or fungi are lost to deeper layers and the water table. In intact forest, the canopy intercepts much of the rainfall and roots recapture leached nutrients before they are permanently lost. When the forest is cleared, this safety net disappears entirely.

> **Buttress Roots:** Large, plank-like roots that extend from the base of tall trees. They spread the enormous weight of the tree over a wider area (since soils are shallow and unable to support deep

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taproots) and also allow the tree to access nutrients near the surface, where decomposition deposits them.

Laterisation: The process by which, following deforestation, the hot sun bakes the exposed, nutrient-depleted soil into a hard, iron-rich crust (laterite). This crust is almost impermeable to water and impossible to cultivate. It can form within a few years of forest clearance and is essentially irreversible on human timescales.

Exam Tip: A key misunderstanding is that rainforest soils are fertile. They are NOT — they are thin and poor. If the forest is cleared, heavy rain leaches away remaining nutrients quickly (laterisation), leaving hard, brick-like soil useless for farming. This is why slash-and-burn farmers must move on after only 2–3 years — the soils are exhausted almost immediately.

Exam Tip — Nutrient Cycle Comparisons: If asked to compare the rainforest nutrient cycle to a temperate deciduous forest, the key differences are: (1) speed — decomposition is far faster in the tropics due to heat and moisture; (2) storage — almost all nutrients are in biomass in the rainforest, whereas temperate soils hold more nutrients; (3) leaching risk — far greater in the rainforest due to higher rainfall intensity.

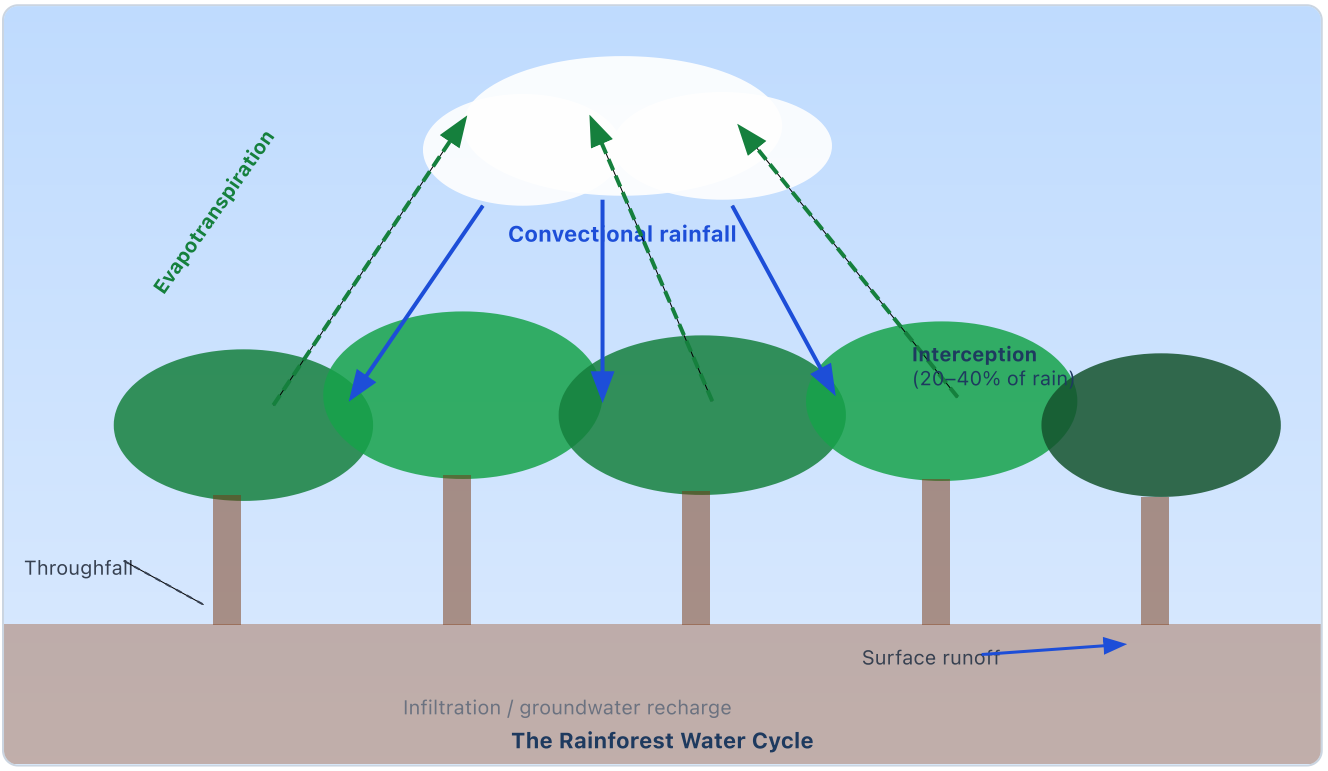
6. The Water Cycle in the Rainforest

The tropical rainforest plays an extraordinary role in the local and global water cycle. The forest does not simply receive rainfall — it actively **generates its own rainfall** through evapotranspiration, creating a self-sustaining feedback loop between vegetation and climate.

Key Processes

Interception: The process by which rainfall is caught by the leaves, branches, and bark of vegetation before it reaches the ground. In a tropical rainforest, the dense multi-layer canopy can intercept 20–40% of all precipitation. This water either evaporates back into the atmosphere directly (interception loss) or drips slowly to the ground as throughfall, greatly reducing the erosive impact of heavy rain on the soil surface.

Evapotranspiration (EVT): The combined process of evaporation (water evaporating from soil and leaf surfaces) and transpiration (water released through the stomata of leaves as a by-product of photosynthesis). In the Amazon, trees release approximately 20 billion tonnes of water vapour into the atmosphere every single day — more than the Amazon River discharges into the Atlantic Ocean. This moisture rises, cools, condenses, and falls again as rainfall — a process sometimes called the "flying rivers" or "forest-rainfall feedback."



The Rainforest Water Cycle

The Forest-Rainfall Feedback Loop

One of the most important — and alarming — aspects of the rainforest water cycle is the **forest-rainfall feedback**. The Amazon generates roughly half of its own rainfall internally through evapotranspiration. Moisture evaporated in the eastern Amazon is carried westward by the trade winds, condenses, falls as rain further west, evaporates again, and is carried still further — creating a "flying river" of moisture that travels thousands of kilometres inland. This means:

- The rainforest is not simply responding to climate — it is **creating the climate it needs to survive**.
- Deforestation at a regional scale interrupts this cycle. Less forest means less evapotranspiration, meaning less moisture in the atmosphere, meaning less rainfall — meaning even more forest dies in a **positive feedback (vicious cycle)**.

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- Research by Antonio Nobre (INPA) suggests that the Amazon generates "flying rivers" carrying up to 20 billion tonnes of water per day westward — more than the Amazon River itself. Cities like São Paulo receive much of their water from this invisible aerial river.

Interception in Detail

The dense multi-layered canopy acts as a giant sponge, cushioning the impact of intense tropical rainfall. Key effects include:

- **Reduced soil erosion:** Rain drops are slowed by leaves and branches before reaching the forest floor, reducing their kinetic energy and the ability to dislodge soil particles.
- **Regulated river flow:** Water reaches rivers more slowly (as throughfall and stemflow rather than direct surface runoff), reducing flood risk downstream.
- **Moisture recycling:** Intercepted water that evaporates directly from leaf surfaces (without entering the soil) feeds local cloud formation and convectional rainfall — keeping the water cycle local and continuous.

> **Exam Tip — Water Cycle Links to Deforestation:** When asked about the effects of deforestation, always include the impact on the water cycle. Without tree cover: (1) interception is lost so rainfall hits bare soil directly, causing erosion and surface runoff; (2) evapotranspiration falls, reducing moisture in the atmosphere and regional rainfall; (3) rivers flood more frequently and more severely. This also links directly to soil degradation and the nutrient cycle.

7. Biodiversity

The tropical rainforest is the **most biodiverse biome on Earth**, home to over 50% of the world's species while covering only about 6% of the land surface.

Why is Biodiversity so High?

- **Stable climate:** Constant warmth and rainfall all year round allows year-round growth and reproduction — no harsh winter or dry season to kill off species.
- **Multiple niches:** The layered structure creates dozens of distinct microhabitats — different species can specialise without competing directly.

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- **Long evolutionary time:** Rainforests have existed for ~60–100 million years without major ice-age disruption, allowing slow accumulation of species through evolution and speciation.
- **High productivity:** Intense photosynthesis produces vast amounts of food energy supporting complex food webs.

Interdependence

Species in the rainforest are **highly interdependent** — the loss of one species can trigger cascading effects throughout the ecosystem. This is sometimes called an ecological cascade or trophic cascade. Key examples:

- **Brazil nut tree (Bertholletia excelsa) — a three-species dependency:** The Brazil nut tree can only be pollinated by the large-bodied **orchid bee** (genus Eulaema), because only this bee is strong enough to lift the heavy petal lid of the flower. The orchid bee itself depends on specific orchid species for a chemical it uses in mating displays — meaning no orchids equals no bees equals no Brazil nuts. Once pollinated, the hard woody seed pod can only be opened by the **agouti**, a large rodent with teeth strong enough to gnaw through the shell. The agouti buries seeds it does not immediately eat — and many of these are never retrieved, becoming new trees. Remove any one of these three species and the whole chain collapses.
- **Fig trees and fig wasps:** Every species of fig tree (over 750 species globally) is pollinated by one or a very small number of dedicated wasp species. The wasp lays eggs inside the fig, the young wasps disperse pollen as they leave. The relationship is so tight that extinction of the wasp means extinction of the fig, and with it the loss of a key food source for dozens of bird and mammal species that depend on fig fruits.
- **Jaguars as apex predators (trophic cascade):** The jaguar controls populations of herbivores including peccaries (wild pigs), capybaras, and deer. Where jaguars have been removed, herbivore populations explode, leading to overgrazing of understorey vegetation and a collapse in plant diversity. This then affects insects, birds, and other species that depend on those plants — demonstrating how removing a single apex predator can restructure an entire ecosystem.
- **Epiphytes and pond-in-the-canopy ecosystems:** Bromeliads (relatives of the pineapple) collect rainwater in their leaf rosettes, forming miniature ponds up to 3 litres in volume high in the canopy. These tiny pools support entire independent ecosystems — specialised mosquito larvae, aquatic beetles, poison dart frog tadpoles, and bacteria found nowhere else. A single large bromeliad may support over 250 animal species. This illustrates how one plant species can create an entire new habitat layer.

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- **Dung beetles and forest regeneration:** Dung beetles bury animal droppings in the soil, dispersing seeds and aerating the soil. Research in the Amazon showed that when large mammals (which produce dung) are removed by hunting, dung beetle populations collapse, seed dispersal is drastically reduced, and forest regeneration slows significantly. This is an invisible form of biodiversity collapse.
- **Strangler figs and forest succession:** Strangler figs germinate high in the canopy as epiphytes, then slowly send roots down to the ground, eventually engulfing and killing the host tree. The dead host leaves a hollow space that becomes critical shelter for bats, insects, and small mammals — turning a process of apparent destruction into habitat creation.

Exam Tip — Interdependence in Extended Answers: For 6–8 mark questions on biodiversity, do not just list species. Pick one or two specific interdependence examples (e.g. the Brazil nut tree chain, or the jaguar trophic cascade) and explain the mechanism clearly: what depends on what, and what happens if the chain is broken. This demonstrates the depth of understanding that separates high from low band answers.

8. Plant Adaptations

Adaptation	What it Does	Why it Matters
Drip tips	Long, pointed leaf tips	Channels rain off the leaf quickly, preventing the growth of mosses and algae that block sunlight and weigh down leaves
Waxy/smooth leaves	Hydrophobic leaf surface	Repels excess water, preventing fungal/bacterial disease in the constantly wet environment
Large leaves	Maximise surface area	Allows understorey plants to capture the tiny amount of light that penetrates the canopy
Buttress roots	Wide plank-like roots at base	Support enormous tree height in shallow soils; access surface nutrients
Epiphytes	Grow on other plants (not parasitic)	Orchids and bromeliads use tree branches to access light in the canopy without needing ground soil

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Lianas	Woody climbing vines	Use trees as support structures to reach canopy light without investing energy in a thick woody trunk
Shallow roots	Roots spread wide near surface	Captures nutrients quickly before they are leached away by rain

Epiphyte: A plant that grows on another plant for physical support but is not parasitic — it obtains its own water (from rain and air moisture) and nutrients (from debris). Examples: orchids, bromeliads, mosses, ferns. Found especially in the canopy layer.

Liana: A woody climbing vine rooted in the soil that uses other trees for support to climb to the canopy. Can grow to over 100m long. Important food source and highway for canopy animals.

9. Animal Adaptations

Animal	Adaptation	Purpose
Toucan	Long, serrated beak (up to 20cm)	Reaches fruit on thin branches that cannot support the bird's weight; also used in social display
Red-eyed tree frog	Adhesive toe pads (suckers); bulging eyes	Grip smooth, wet leaves/branches in canopy; large eyes for low-light understorey conditions
Jaguar	Spotted coat (dappled pattern)	Camouflage in the broken light of the forest floor; ambush predator
Sloth	Extremely slow movement; low metabolic rate; stomach bacteria to digest leaves	Leaves have very low nutritional value; slow movement conserves energy and makes the sloth hard for predators to spot
Poison dart frog	Bright warning colouration (aposematism); skin toxins derived	Warns predators of toxicity; the toxins themselves come from their diet of specific

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	from their diet of ants/mites	insects found only in the rainforest
Flying squirrel / gliding animals	Skin membrane between limbs	Glides between trees — efficient travel in the dense forest canopy without descending to the dangerous forest floor

10. Deforestation

Deforestation: The permanent removal of trees from a forested area, converting land to non-forest use. The global rate is approximately **10 million hectares per year** (equivalent to losing a football pitch every 2–3 seconds).

Causes of Deforestation

Cause	Detail	Scale / Significance
Cattle ranching	Forest cleared for pasture — cheapest land use	Biggest single driver in the Amazon (most cleared land becomes cattle pasture)
Soya farming	Global demand for soya (livestock feed, biofuel)	Brazil is world's largest soya exporter; the 2006 Soy Moratorium cut direct clearance for soya in the Amazon, but soya expansion still pushes ranchers deeper into the forest
Commercial logging	Hardwoods (mahogany, teak) sold internationally	Legal and illegal logging opens up roads into intact forest
Mining	Iron ore, gold, bauxite extraction — Amazon has vast mineral wealth	Creates pollution, mercury contamination of rivers (illegal gold mining)
HEP dams	Large reservoirs flood vast forest areas	Belo Monte Dam (Brazil) flooded 500km² of Amazon

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Road building	Trans-Amazonian Highway opened up interior	Enables further settlement, farming, logging along new routes
Population growth	Land hunger — small farmers clear forest for subsistence	Migration to Amazon from overcrowded cities; landlessness
National debt	Developing countries exploit forests for export revenue to repay debt	Short-term economic pressure overrides environmental concerns

Consequences of Deforestation — in Detail

1. Biodiversity Loss

Habitat destruction is the single greatest driver of species extinction globally. Some scientists have estimated that as many as **137 plant, animal and insect species are lost every day** from rainforests worldwide — roughly one species every ten minutes. Because of the high interdependence of rainforest species, the extinction of one species can trigger a cascade of secondary extinctions. A 2015 study estimated that between a third and a half of Amazonian tree species could qualify as threatened with extinction because of deforestation. Many species are lost before they are even discovered or described by science.

2. Soil Erosion and Degradation

Without root systems binding the soil and leaf litter cushioning the impact of rain, intense tropical downpours dislodge and wash away topsoil at rates **up to 100 times faster** than in intact forest. The exposed laterite soil rapidly loses structure and fertility. Within 2–3 agricultural cycles, the land becomes essentially unusable for farming — yet it may take centuries for natural forest to regrow on degraded land. It has been estimated that around 2 billion hectares of deforested or degraded land worldwide could be restored.

3. Disruption of the Water Cycle

Trees recycle **50–80% of rainfall** through evapotranspiration. When a large area is deforested, regional rainfall can decline significantly. Studies suggest that heavily deforested parts of the Amazon already receive **noticeably less rainfall** and have longer dry seasons than forested areas. This creates a positive feedback: less forest = less rain = more drought stress on remaining trees = more forest die-off. Cities far from the Amazon, including São Paulo (population 22 million), receive significant proportions

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of their rainfall from Amazonian "flying rivers" — meaning deforestation in Brazil creates water shortages in cities hundreds of kilometres away.

4. Climate Change — Carbon Release

Tropical forests store approximately **250 billion tonnes of carbon** in their biomass and soils — equivalent to about 25 years of current global CO_2 emissions from fossil fuels. When forest is burned or cleared, this carbon is released as CO_2 and methane. Deforestation and land-use change currently account for approximately **10–15% of annual global greenhouse gas emissions**. Crucially, a degraded Amazon could transition from a carbon sink (absorbing CO_2) to a carbon source (emitting CO_2) — research published in 2021 in Nature showed that the south-eastern Amazon is already emitting more carbon than it absorbs, due to a combination of deforestation and climate-driven drought.

5. Displacement and Rights of Indigenous People

The Amazon is home to **hundreds of Indigenous peoples** (well over 1 million people), including dozens of uncontacted groups. Deforestation destroys not only their physical environment but their culture, traditional knowledge (including plant-based medicines), and survival systems. The Yanomami crisis of 2022–23, when 20,000 illegal gold miners (garimpeiros) invaded Yanomami territory, resulted in widespread malaria, mercury poisoning, malnutrition, and a dramatic rise in child mortality. The Brazilian health authority declared a public health emergency in January 2023.

6. Loss of Medicinal Potential

An estimated **25% of all modern pharmaceutical drugs** are derived from or inspired by rainforest plants. The muscle relaxant curare (used in anaesthesia) comes from Amazonian plants; quinine (malaria treatment) from Andean cinchona bark; and vincristine (childhood leukaemia treatment) from the rosy periwinkle of Madagascar. Given that less than 1% of known rainforest plant species have been scientifically tested for medicinal properties, the pharmaceutical potential that is being permanently destroyed is incalculable.

Exam Tip — Using Statistics Effectively: In 6–8 mark questions, examiners reward specific factual support. Always try to include at least one statistic per consequence you describe. Do not invent statistics — use the ones you have memorised (up to 137 species/day; 250 billion tonnes carbon; 25% of medicines from rainforest plants). Quoting a study or named scientist (e.g. "research published in Nature, 2021") adds further credibility.

Deforestation and economic development — the other side

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Deforestation is not only negative. For countries such as Brazil and Malaysia it has brought **economic development**:

- Jobs in logging, farming, mining and processing — the Brazilian beef and soya industries employ millions of people.
- Export income — Brazil is the world's largest exporter of beef and soya; Malaysia and Indonesia produce around 85% of the world's palm oil.
- Taxes from these industries pay for roads, schools and hospitals; HEP dams such as Belo Monte provide electricity.
- However, profits often go to large companies and wealthy landowners, and the long-term costs (soil erosion, lost tourism and ecosystem services, climate change) may outweigh short-term gains.

11. Case Studies

Case Study A: The Amazon Rainforest, Brazil

Location: South America, spanning 9 countries. Approximately 60% lies within Brazil. The Amazon River drains the basin — the world's largest river by discharge volume.

Size: ~5.5 million km² of tropical rainforest — roughly the size of Western Europe.

Biodiversity: Home to an estimated 10% of all Earth's species: 40,000 plant species, 1,300 bird species, 3,000 fish species, 430 mammal species.

Rate of loss: Approximately 17% of the forest has been cleared since the 1970s. Peak deforestation was in 2004 (~27,000 km²/year). After falling following policy intervention (Forest Code), it rose again after 2018 — under Bolsonaro's government (2019–2022), deforestation reached 10,000–13,000 km²/year (peaking at about 13,000 km² in 2021). Under Lula (from 2023), deforestation dropped significantly again — to around 6,000 km² in 2024, roughly half the 2021 peak.

Main causes in the Amazon: Cattle ranching (the largest driver), soya cultivation (often indirectly), logging, mining (particularly illegal gold mining in indigenous territories), road building.

Tipping Point: Scientists (particularly Carlos Nobre and Thomas Lovejoy) warn that if 20–25% of the Amazon is deforested, the forest may reach a 'tipping point' — reduced evapotranspiration

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causes regional drying that prevents forest regrowth. As of 2024, approximately 17–20% has been cleared and a further ~17% is degraded. Seasonal drought events driven by climate change have worsened — in 2023, an extreme drought reduced water levels on the Amazon River to historic lows, stranding river communities and killing more than 150 river dolphins in Lake Tefé; the 2024 drought brought even lower river levels.

Indigenous impacts: The Yanomami people (35,000 individuals) suffered a major humanitarian crisis in 2022–23 when 20,000 illegal gold miners (garimpeiros) invaded their territory, spreading malaria and mercury poisoning. Child deaths from malnutrition and disease rose sharply, and Brazil declared a public health emergency in January 2023.

Significance: The Amazon stores approximately 150–200 billion tonnes of carbon. Its loss would accelerate global climate change significantly and is considered by scientists to be one of the most critical planetary tipping points.

Case Study B: Borneo (SE Asia) — Deforestation for Palm Oil

Location: Borneo is the world's third largest island, divided between Indonesia (Kalimantan), Malaysia (Sabah and Sarawak), and Brunei. It lies roughly between 7°N and 4°S, straddling the equator, in the heart of the Indo-Malayan rainforest region.

Size of original forest: Borneo (about 740,000 km²) was once almost entirely covered by rainforest. It is home to some of the oldest rainforest on Earth — estimated at 130 million years old, significantly older than the Amazon.

Rate of loss: Between 1973 and 2010, Borneo lost approximately **30% of its forest cover** — about 170,000 km² (larger than England). Deforestation in the 2000s was one of the fastest rates on Earth at the time. The primary driver has been **palm oil plantations**, which now cover over 6 million hectares in Malaysian and Indonesian Borneo combined.

Main causes:

- **Palm oil:** Used in ~50% of all supermarket products globally (biscuits, shampoo, margarine, cosmetics). Palm oil is the highest-yielding vegetable oil and the cheapest to produce, making it extremely profitable to grow. Vast areas of peatland rainforest have been drained and burned to make way for plantations.
- **Commercial logging:** Tropical hardwoods (meranti, teak) exported primarily to Japan, China, and Europe. Both legal and illegal logging have been major drivers since the 1960s.

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- **Mining:** Coal and bauxite mining in Kalimantan (Indonesian Borneo).
- **Transmigration:** The Indonesian government's policy of relocating people from densely populated Java to Borneo, placing pressure on land from subsistence farming.

Ecological consequences:

- **Orangutan:** The Bornean orangutan (Pongo pygmaeus) lost nearly **150,000 individuals between 1999 and 2015** — roughly half the population — primarily due to habitat loss and hunting. It is now classified as Critically Endangered (IUCN). Orangutans need large contiguous forest territories and cannot survive in plantation monocultures.
- **Bornean pygmy elephant and Sumatran rhino:** the elephant is Endangered and the rhino Critically Endangered (now extinct in the wild in Malaysian Borneo), with small fragmented populations increasingly unable to maintain viable breeding populations.
- **Peat fires:** Borneo's peatlands (carbon-rich waterlogged soils up to 20m deep) have been drained for plantations. When drained peat dries out, it becomes highly flammable. The 1997–98 El Niño-driven fires in Indonesia burned roughly 10 million hectares of land (much of it peat), releasing an estimated 0.8–2.6 billion tonnes of carbon — equivalent to 13–40% of global fossil fuel emissions for that year. Similar large-scale fires occurred in 2015 and 2019.
- **River pollution:** Palm oil mill effluent (POME) has severely polluted rivers in Sabah and Kalimantan, with dissolved oxygen levels in some rivers falling to near zero, wiping out freshwater fish populations.

Responses: The Roundtable on Sustainable Palm Oil (RSPO) was established in 2004 to certify sustainably produced palm oil. By 2023, approximately 19% of global palm oil was RSPO-certified. Critics argue the RSPO's standards are insufficiently strong and enforcement is weak. Some countries (Norway, UK) have committed to "deforestation-free" supply chains. Consumer pressure has led some companies (Unilever, Nestle) to commit to sourcing only certified palm oil — but compliance is difficult to verify across complex global supply chains.

Comparison with Amazon: Borneo's deforestation is driven more by a single global commodity (palm oil) than the Amazon's multiple drivers (cattle, soya, logging, mining). This potentially makes it easier to address through supply chain certification — but the global scale of palm oil use makes systemic change extremely challenging. Borneo's forests are also unique — they share very few species with the Amazon and are irreplaceable even if the Amazon were fully protected.

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Exam Tip — Using Two Case Studies: If asked to "use a case study to support your answer," one well-chosen example with specific facts scores higher than a vague general statement. If the question says "with reference to examples," you can briefly compare both case studies — e.g. "In the Amazon, the main driver is cattle ranching, whereas in Borneo, palm oil for global consumer goods is the primary driver, affecting over 6 million hectares." Contrast shows geographical understanding of scale and variation.

12. Climate Regulation — Global Importance of Rainforests

Rainforests are not merely local or regional ecosystems — they perform **critical services for the entire planet's climate system**. Their global importance extends far beyond the countries in which they are physically located.

Carbon Storage and the Carbon Sink Function

Tropical rainforests together store approximately **250 billion tonnes of carbon** — roughly equivalent to 25 years of current global CO_2 emissions from all fossil fuel burning combined. This carbon is held in:

- **Living biomass** (above ground: trunks, branches, leaves; below ground: roots)
- **Dead wood** (fallen logs, bark, litter layer)
- **Soil organic matter** (particularly in peat-rich soils like those of Borneo)

When trees photosynthesise, they absorb CO_2 from the atmosphere and lock it into woody tissue — acting as a **carbon sink**. Intact tropical forests currently absorb a net 1.1–1.4 billion tonnes of carbon per year, significantly offsetting global emissions. However, this sink is weakening: a 2021 study in Nature found that heavily deforested and degraded parts of the Amazon (particularly in the south-east) have flipped to become net carbon **sources**, emitting more than they absorb. If this transition continues across the whole Amazon system, it would dramatically accelerate global warming.

Albedo and Surface Temperature Regulation

Rainforest canopies are relatively dark and absorb more solar radiation than bare or agricultural land. Counterintuitively, forest also causes significant **evaporative cooling** — the enormous volumes of water transpired by leaves absorb latent heat, cooling the local air temperature. Studies comparing deforested and forested areas in the tropics show that deforested land is on average **1–3°C warmer** than

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equivalent forested land. At a regional scale, large-scale deforestation warms regional temperatures, worsening drought conditions and potentially affecting atmospheric circulation patterns.

Influence on Global Weather Patterns

The Amazon acts as a major driver of atmospheric circulation. The enormous amounts of water vapour and heat released by evapotranspiration drive convection currents that influence the **position and strength of the jet stream** and inter-tropical weather systems. Climate modelling studies suggest that complete loss of the Amazon would significantly alter rainfall patterns as far away as the **Midwestern United States and parts of Europe** — demonstrating that the rainforest's climate services are genuinely global in reach, not merely regional.

Biodiversity as a Climate Stabiliser

High biodiversity itself contributes to ecosystem resilience and stability. A more diverse forest is better able to withstand and recover from disturbances such as drought, disease, or insect outbreaks. As climate change intensifies, maintaining high biodiversity in the rainforest may be critical to its ability to continue performing its climate regulation functions — a further reason why biodiversity loss and climate change are interlinked crises, not separate problems.

Carbon Sink: A natural system that absorbs more carbon (as CO_2) from the atmosphere than it releases. Intact tropical rainforests are major carbon sinks. When deforested, they become carbon sources — releasing stored carbon as CO_2.

Ecosystem Services: The benefits that healthy ecosystems provide to people, free of charge. For rainforests, key services include: carbon storage (climate regulation), water cycle regulation, biodiversity (gene pool for medicine and agriculture), local rainfall generation, and soil protection. These services are estimated to be worth trillions of dollars globally if costed — yet they are provided by nature without payment.

Exam Tip — Why Should Wealthy Countries Care? A common exam question asks why people in the UK should care about deforestation in Brazil or Indonesia. Key arguments: (1) CO_2 released anywhere affects global climate equally; (2) UK consumers drive demand for beef, soya, and palm oil that cause deforestation; (3) Rainforest biodiversity may hold future medical cures; (4)

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atmospheric circulation means Amazonian deforestation may affect European weather. This demonstrates understanding of globalisation, interdependence, and global responsibility.

13. Sustainable Management

Sustainable management aims to use the rainforest to meet present needs **without compromising the ability of future generations** to benefit from it.

Strategy	How it Works	Evaluation
Selective logging	Only mature trees of specific species are felled; the rest of the forest is left intact	Reduces biodiversity loss vs. clear-felling; but roads still open up forest; takes 50+ years for full recovery
Replanting / reforestation	New trees planted to replace those felled; often monocultures	Restores carbon storage; but monocultures have far less biodiversity than natural forest
Ecotourism	Tourists pay to visit and experience the forest; income supports conservation	Costa Rica: 26% of land protected; ecotourism worth $3bn/year. However: can disturb wildlife; limited to accessible areas
Debt-for-nature swaps	A country's debt is reduced or cancelled in exchange for committing to protect a specified area of forest	Bolivia, Ecuador used this model. Effective when government has capacity to enforce; corruption can undermine it
National parks and reserves	Legal protection from development (e.g. Tumucumaque National Park, Brazil — world's largest tropical national park)	Effective if well-enforced; illegal logging/mining still occurs in many "protected" areas
CITES	International agreement banning trade in endangered species and protected timber (e.g. mahogany)	Reduces commercial incentive to log protected species; but enforcement difficult

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Indigenous land rights	Legal recognition of indigenous peoples' rights to their traditional territories — they are proven best forest guardians	Indigenous territories in the Amazon have significantly lower deforestation rates; but politically contested
REDD+	UN scheme: developed countries pay developing countries to protect and restore forests as carbon offsets	$billion-scale funding available; complex governance challenges; some criticism of "carbon offsetting" as greenwashing

Deeper Evaluation of Key Strategies

Indigenous Land Rights — the strongest evidence: Research comparing deforestation rates across different land tenure types in the Amazon consistently shows that indigenous territories have **dramatically lower deforestation rates** than equivalent non-protected forest or even formally designated national parks. Studies of satellite data have found that indigenous lands in Brazil have **much lower deforestation rates** than non-protected areas. Indigenous communities have the greatest incentive to protect their territories and have accumulated millennia of ecological knowledge about sustainable forest use. Yet these rights are frequently contested by governments and extractive industries — and under Bolsonaro, demarcation of new indigenous territories was halted entirely, directly contributing to increased deforestation.

REDD+ — promising but flawed: REDD+ (Reducing Emissions from Deforestation and Forest Degradation) is the most ambitious international mechanism, theoretically capable of channelling hundreds of billions of dollars from wealthy countries into forest protection. However, evaluation reveals significant weaknesses: (1) **additionality** — is the forest only protected because of REDD+ money, or would it have survived anyway? (2) **permanence** — carbon offsets from protected forest can be undermined if the forest is later burned; (3) **leakage** — protecting forest in one area may simply displace deforestation elsewhere; (4) a 2023 investigation found that over 90% of rainforest carbon credits issued by one major certifier (Verra) were "phantom credits" that did not represent real emissions reductions. This does not invalidate REDD+ entirely, but shows that robust monitoring and governance are essential.

Ecotourism — effective but limited scale: Costa Rica remains the most cited success story: forest cover recovered from 21% (1987) to over 52% (2023) and ecotourism contributes over $3 billion/year to GDP. However, Costa Rica is a small, stable, middle-income country with good governance. Attempting to replicate this in the DRC (Congo Basin) — a larger, lower-income country with fragile governance and

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ongoing conflict — is far more challenging. Ecotourism also inherently requires visitor access, which limits its applicability to the most remote and ecologically valuable areas of the Amazon or Congo.

Consumer action and supply chains: Since most Amazon deforestation is driven by beef and soya production for global markets, consumer and corporate behaviour in importing countries can be powerful. The UK's Environment Act 2021 contains provisions requiring large companies to ensure their supply chains are free from illegal deforestation — one of the first such laws globally. The EU Deforestation Regulation (EUDR, agreed in 2023 but its start date has been postponed more than once) goes further, requiring companies to prove their products (beef, soya, palm oil, coffee, cocoa, rubber, timber) did not cause deforestation before they can be sold in the EU. These laws, if well-enforced, could significantly reduce the economic incentive for deforestation by cutting off market access for destructive producers.

Case Study: Ecotourism in Costa Rica

Costa Rica has transformed from one of the world's highest deforestation rates in the 1970s to a global leader in conservation. Today, 26% of its land area is protected. Ecotourism brings in over $3 billion per year and employs 13% of the workforce. The government's **Payment for Ecosystem Services (PES)** scheme pays landowners to keep forest standing. Forest cover has recovered from 21% (1987) to over 52% (2023). This demonstrates that economic incentives aligned with conservation can be more effective than protection alone.

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